TOPIC 2 Transforming and Comparing Linear Functions

Name _____ Date _____

1. What are the equation and slope of the vertical line through (−7, 2)?

A. y = 2; slope is zero

B. x = −7; slope is zero

C. x = −7; slope is undefined

D. y = −7; slope is undefined

Vertical: every point has x = −7. Slope = rise ÷ 0: undefined (VUX).

2. Write the equation in slope-intercept form of the line through (−1, 0) that is parallel to $y = 5x - 2$.

***y* = 5*x* + 5**

Same slope 5. 0 = 5(−1) + b → b = 5.

3. $f(x) = 2x + 1$ is translated 3 units left. What is the y-intercept of the new function?

7

g(x) = f(x + 3) = 2(x + 3) + 1 = 2x + 7. y-intercept 7.